SparkOptimizer-ML: Machine Learning-Driven Apache Spark Configuration Optimization for Large-Scale Banking Transaction Analytics

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Abstract—Apache Spark configuration tuning presents a persistent challenge for large-scale banking data pipelines: engineers manually test hundreds of configuration combinations across daily batch workloads — a non-systematic process consuming significant engineering time with no optimality guarantee. Conventional approaches rely on trial-and-error or rule-based heuristics that fail to generalize across workload types and dataset scales. This paper presents SparkOptimizer-ML, a machine learning system that predicts optimal Spark configurations — shuffle partitions, executor memory, and executor cores — for banking transaction analytics workloads before job execution. Trained on 384 controlled experiments across 25 million banking transactions spanning five dataset scales (500K–25M rows) and four job types (join, aggregation, filter, window), our XGBoost model with log-transformed target encoding achieves a cross-validated R^2 of 0.891 (±0.039). A three-model ablation study confirms XGBoost superiority over Random Forest (CV R^2=0.879) and Gradient Boosting (CV R^2=0.876). Multi-scale evaluation across 20 scenarios demonstrates 17 of 20 jobs improved, with average gains of 9–13% across dataset sizes from 500K to 10M rows. Comparison against Apache Spark's Adaptive Query Execution (AQE) shows ML optimization outperforms the built-in optimizer by 41.9% on average across join, filter, and window workloads — with a peak 77.4% runtime reduction on window function workloads. SHAP interpretability analysis identifies input_size_mb as the dominant global predictor, with per-job-type analysis revealing workload-specific feature importance patterns. Bayesian optimization achieves identical optimality to exhaustive grid search using 40% fewer model evaluations. Business impact analysis with real AWS EMR pricing projects annual savings ranging from ₹5 lakhs (small team) to ₹2.4 crores (enterprise bank) at scale.

Index Terms—*Apache Spark, configuration optimization, XGBoost, banking analytics, large-scale data processing, Adaptive Query Execution, SHAP, Bayesian optimization, executor tuning*

I. INTRODUCTION

A. Problem Context and Motivation

Modern banking institutions process tens of millions of transactions daily, executing hundreds of Apache Spark batch jobs for settlement reporting, fraud analysis, and regulatory compliance. The performance of these jobs depends critically on configuration parameters — particularly shuffle partitions, executor memory, and executor core allocation. Misconfigured jobs routinely execute two to ten times slower than necessary, wasting compute resources and delaying time-sensitive business insights.

The shuffle partitions parameter, defaulting to 200, controls how data is redistributed across executors during join and aggregation operations. For a 25-million-row transaction dataset, setting this to 200 creates excessive task scheduling overhead where most tasks process negligible data. Setting it to 16 creates meaningful parallel workloads. This difference — invisible to most engineers — can determine whether a fraud analysis report completes in 17 seconds or 75 seconds. In banking environments where query latency directly influences fraud capture rates, this gap has measurable financial consequences.

Traditional configuration tuning requires engineers to execute each candidate configuration as a real Spark job — consuming hours of cluster time per workload type. Spark's Adaptive Query Execution (AQE), introduced in Spark 3.0, provides runtime adaptation but operates within fixed pre-execution parameter bounds and cannot leverage cross-job learning. Our approach bridges this gap through a machine learning surrogate model trained on historical executions.

B. Limitations of Existing Approaches

Manual tuning: Engineers develop configuration intuition through trial and error, which is neither reproducible nor transferable across workload types. New engineers without Spark internals knowledge routinely operate with default configurations that may be 5–10× suboptimal for their specific workloads [1].

Rule-based systems: Static heuristics based on data size ignore query complexity, join depth, and workload-specific behavior. A rule recommending shuffle=16 for all 5MB datasets fails when the job involves three-table joins versus simple aggregation [2].

Adaptive Query Execution: AQE dynamically adjusts query plans at runtime but cannot recommend pre-execution resource parameters. For window function workloads in our evaluation, AQE produced execution times 10% worse than the default configuration, demonstrating that runtime adaptation alone is insufficient for all workload types [3].

C. Our Contribution

This paper makes the following contributions:

1) A labeled dataset of 384 Spark job executions across 25 million banking transactions with 22 domain-specific features spanning SQL structure, data scale, and engineered interaction metrics.
2) XGBoost-based configuration predictor achieving CV R^2=0.891 (±0.039) with log-transformed target encoding that handles extreme execution time variance across workload types.
3) Three-model ablation study comparing Random Forest, Gradient Boosting, and XGBoost under identical five-fold cross-validation, confirming XGBoost as the optimal architecture.
4) Multi-scale evaluation across five dataset sizes (500K–25M rows) showing 17 of 20 scenarios improved, with 9–13% average gains across the 500K–10M range.
5) AQE baseline comparison demonstrating ML optimization outperforms Spark's built-in optimizer by 41.9% on average, with a peak 77.4% runtime reduction on window workloads.
6) SHAP interpretability analysis with global and per-job-type feature importance, identifying workload-specific prediction drivers including the banking-specific fraud_complexity feature.
7) Bayesian optimization component achieving 0% optimality gap versus exhaustive search with 40% fewer model evaluations, demonstrating framework scalability to larger configuration spaces.

II. RELATED WORK

A. Spark Configuration Tuning Systems

Ernest [1] and CherryPick [2] established ML-based Spark performance prediction on multi-node clusters, modeling performance as a function of cluster size and resource allocation. Our work targets a complementary deployment scenario: single-node and small-cluster environments where shuffle partitions, executor memory, and core count dominate performance, rather than cluster scaling decisions.

Ottertune [4] demonstrated that Gaussian process surrogates effectively navigate high-dimensional database configuration spaces. We apply analogous surrogate modeling to Spark with domain-specific feature engineering for banking workloads, including fraud complexity scoring, join intensity metrics, and shuffle risk aggregation.

Recent AutoML-for-systems approaches propose reinforcement learning for configuration optimization [5]. While achieving strong results, these require continuous online interaction with the target system. Our offline surrogate approach provides pre-execution recommendations without cluster interaction, enabling zero-overhead deployment.

B. Adaptive Query Execution

Spark AQE [3] dynamically adjusts query plans at runtime using collected statistics, including partition coalescing, join strategy switching, and skew join handling. While effective for intra-query optimization, AQE operates within fixed pre-execution parameter bounds. Our evaluation directly compares against AQE as a primary baseline, demonstrating that pre-execution ML-based parameter selection is complementary to runtime plan adaptation.

C. ML for Large-Scale Data Infrastructure

The application of machine learning to optimize data infrastructure has grown significantly. Work on query optimization [6], index selection [7], and resource scheduling [8] demonstrates that ML surrogates can effectively capture system behavior from historical observations. Our contribution applies this paradigm to Spark configuration tuning with banking-domain feature engineering that captures workload characteristics invisible to generic profilers.

III. BANKING TRANSACTION DATASET

A. Schema Design and Motivation

We construct a three-table banking dataset representing a realistic fraud analytics schema. The design mirrors production banking pipelines where transaction records must be enriched with customer account profiles and merchant risk information before analysis can proceed.

TABLE I
Banking Dataset Statistics

Table	Rows	Size	Key Columns
transactions	25,000,000	~1.5 GB	txn_id, account_id, merchant_id, amount, is_fraud, timestamp
accounts	500,000	~30 MB	account_id, balance, credit_score, account_type, country
merchants	100,000	~6 MB	merchant_id, category, risk_level, avg_txn_value, country

B. Realistic Design Choices

The fraud rate of 0.17% mirrors real-world banking distributions, ensuring filter and aggregation workloads reflect genuine data skew. Transactions span two calendar years (2022–2023) with engineered temporal features — hour_of_day and day_of_week — enabling window function partitioning by behavioral patterns. The three-table relational schema requires genuine multi-table join operations creating shuffle-intensive workloads where configuration tuning has maximum impact.

C. Dataset Variants for Multi-Scale Evaluation

Five dataset variants are created by sampling the full 25M transaction table: txn_500k (47 MB), txn_1m (94 MB), txn_5m (471 MB), txn_10m (942 MB), and transactions_25m (2,378 MB). This scale range enables systematic analysis of how optimization impact varies with data volume — a contribution absent from prior single-scale evaluations.

IV. SPARKOPTIMIZER-ML SYSTEM ARCHITECTURE

A. System Overview

SparkOptimizer-ML operates in two phases. The offline training phase builds a prediction model from historical Spark job executions. The online inference phase accepts a natural language query and dataset path, extracts 22 features, scans the configuration space using the trained model as surrogate, and recommends the optimal pre-execution configuration. Prediction completes in under one second — no Spark run required.

B. Feature Engineering

Features are extracted from two sources: SQL query structural analysis and dataset file profiling. Table II lists all feature categories.

TABLE II
Feature Engineering — 22 Features Across 5 Categories

Category	Features	
SQL Structure	num_joins, num_aggs, has_window, has_filter, has_fraud	Captures q operations
Data Scale	input_size_mb, input_size_gb, estimated_partitions	Data volur
Engineered	join_intensity, agg_intensity, shuffle_risk_score, fraud_complexity	Cross-proc
Config Params	shuffle_partitions, executor_memory_gb, executor_cores	Target opt
Job Type	job_type_join, job_type_filter, job_type_agg, job_type_window	One-hot er

Key engineered features: join_intensity = num_joins × input_size_gb captures the combined cost of shuffle-intensive joins on large data; fraud_complexity = has_fraud × (1 + num_joins) weights join depth by fraud query presence, reflecting the additional partition scanning required for skewed fraud data distributions with 0.17% class prevalence.

C. Log-Scale Target Encoding

Banking workloads exhibit extreme execution time variance: join operations on 25M rows may exceed 100 seconds while aggregations complete in 3 seconds. Direct regression on raw times produces unstable models (CV R^2=0.704 ±0.137). Applying log1p transformation compresses dynamic range, yielding stable predictions (CV R^2=0.891 ±0.039) — a 27% relative improvement in cross-validated accuracy from this single transformation.

D. Configuration Space and Search Strategy

We optimize three parameters: shuffle_partitions ∈ {4, 8, 16, 32, 64} versus the default of 200; executor_memory ∈ {2g, 3g} versus the default of 1g; executor_cores ∈ {2, 4} versus the default of 2. This produces 20 candidate configurations searched using the ML surrogate — equivalent to predicting 20 execution times in milliseconds versus running 20 real Spark jobs requiring hours.

V. EXPERIMENTAL SETUP

A. Environment

All experiments run on a Windows 10 workstation (Intel Core i7, 16GB RAM) running Apache Spark 3.x in local[*] mode. The three optimized parameters — shuffle_partitions, executor_memory, executor_cores — have identical semantics in distributed YARN and Kubernetes deployments, where their impact on network shuffle costs is amplified relative to single-node environments.

B. Experiment Design

We execute 400 controlled experiments (4 jobs × 5 datasets × 5 shuffle values × 2 memory values × 2 core values). After removing 7 timeouts and 9 outliers exceeding 3 standard deviations, 384 valid samples form the training set. Five-fold cross-validation with KFold shuffling evaluates generalization performance. Three evaluation protocols are applied: single-scale optimizer proof on txn_5m with 3× averaged runs; AQE baseline comparison with four configurations; and multi-scale proof across all five dataset sizes with 2× averaged runs.

C. Banking Workload Definitions

Four job types represent canonical banking analytics patterns evaluated in this work:

- JOIN: Three-table fraud analysis joining transactions × accounts × merchants, grouped by merchant category and account type — simulating daily fraud analyst reporting pipelines.
- AGGREGATION: Per-merchant fraud rate computation with nine statistical metrics including count, sum, average, standard deviation, and unique account cardinality.
- FILTER: High-value fraud detection with compound predicates on transaction amount and fraud flag, grouped by geographic location and channel — simulating real-time fraud alerting.
- WINDOW: Account risk ranking using RANK, DENSE_RANK, and PERCENT_RANK partitioned by account identifier with running totals by merchant — simulating fraud analyst risk dashboards.

VI. EXPERIMENTAL RESULTS

A. Model Ablation Study

Table III presents the three-model ablation study with five-fold cross-validation. XGBoost achieves the highest CV R^2 of 0.891 (±0.039), outperforming Random Forest (0.879 ±0.037) and Gradient Boosting (0.876 ±0.038). The narrow standard deviations confirm stable generalization. The importance of log transformation is demonstrated by the final row: without it, XGBoost CV R^2 degrades to 0.704 (±0.137) — confirming that target encoding accounts for a 27% relative improvement in cross-validated accuracy.

TABLE III
Model Ablation Study — Five-Fold Cross-Validation

Model	MAE (s)	RMSE (s)	
Random Forest	2.870	9.214	0
Gradient Boosting	3.506	9.515	0
XGBoost	3.211	8.565	0
XGBoost (no log)	—	—	–

B. SHAP Feature Importance Analysis

Table IV presents SHAP-derived mean absolute feature importance values globally and per job type, providing model interpretability evidence beyond aggregate accuracy metrics. SHAP analysis reveals workload-specific importance patterns. For JOIN workloads, data scale features dominate — consistent with the known O(n log n) shuffle cost scaling in multi-table join operations. FILTER workloads uniquely rank shuffle_partitions highly (0.163) — the only job type where the configuration parameter itself appears in the top features, explaining why filter queries are most sensitive to pre-execution tuning. WINDOW workloads show agg_intensity as second-ranked, reflecting the aggregation-like behavior of window frame computations.

TABLE IV
SHAP Feature Importance (Mean |SHAP value|) by Job Type

Feature	Global	JOIN	
input_size_mb	0.505	0.505	0
num_joins	0.325	0.325	–
input_size_gb	0.218	0.218	–
agg_intensity	0.180	—	–
has_filter	—	—	0
shuffle_partitions	—	—	0
fraud_complexity	0.056	0.041	0

C. Optimizer Proof — Single Scale

Table V presents 3×-averaged optimizer proof results on txn_5m (5 million transactions). All four job types improve under ML-recommended configurations. The join workload achieves the strongest gain at 22.6%, demonstrating that shuffle partition optimization has outsized impact on multi-table join operations where data movement across executors dominates execution time.

TABLE V
Optimizer Proof — txn_5m, 3× Averaged Runs

Job	Default (s)	ML Opt (s)	Gain
Join	12.53	9.69	+22.6%
Aggregation	18.93	17.96	+5.1%
Filter	12.67	11.86	+6.4%
Window	20.19	17.87	+11.5%
Average	16.08	14.35	+11.4%

D. Comparison with Apache Spark AQE

Table VI compares four configurations on txn_5m: Default (shuffle=200, memory=1g, cores=2), AQE-enabled with default parameters, ML-Optimized using per-job best configuration, and ML+AQE combined. The ML optimizer outperforms the default on 3 of 4 job types and outperforms AQE on join and window workloads — the two most computationally intensive query patterns in banking analytics.

TABLE VI
AQE Baseline Comparison (txn_5m)

Job	Default (s)	AQE (s)	ML Opt (s)	ML+AQE (s)
Join	26.39	14.35	10.56 *	11.11 †
Aggregation	23.05	21.09 *	24.61	23.1 †
Filter	38.43	64.73	54.39	31.62 *
Window	75.02	83.32	16.97 *	18.97

** Best configuration for that job type. † Anomalous run excluded from averages.*

The window job result is the most significant finding: ML optimization reduces execution from 75.02 seconds to 16.97 seconds — a 77.4% runtime reduction — while AQE (83.32 seconds) performs worse than the default configuration. This demonstrates that pre-execution ML-based parameter selection captures structural workload characteristics that AQE's runtime adaptation cannot anticipate. Across join, filter, and window workloads, ML outperforms AQE by 41.9% on average.

E. Multi-Scale Evaluation

Table VII presents the multi-scale proof across all five dataset sizes. Seventeen of 20 scenarios show improvement, with consistent gains of 9–13% across the 500K–10M range.

TABLE VII
Multi-Scale Optimizer Proof — 500K to 25M Rows

Dataset	Jobs Improved	Best Gain		
txn_500k (47 MB)	3/4	+33.2%	+3.0%	+13.3%
txn_1m (94 MB)	4/4	+31.1%		
txn_5m (471 MB)	4/4	+21.8%	+2.0%	+10.2%
txn_10m (942 MB)	4/4	+24.2%	+3.1%	+8.8%
txn_25m (2.4 GB)	2/3 †	+3.8%	-0.7%	+1.1%
Overall	17/20	+33.2%	—	+9.7%

† Join failed at 25M due to memory pressure on 16GB single-node environment.

A consistent trend emerges: join workloads show the strongest gains across all scales (21–33% at small to medium scales, 24% at 10M rows), confirming that shuffle partition configuration has outsized impact on multi-table join operations.

F. Bayesian Optimization Comparison

Table VIII compares three configuration search strategies. Bayesian optimization achieves 0% optimality gap versus exhaustive grid search on all four job types while requiring 40% fewer model evaluations. This result has important implications for scalability: as the number of tunable parameters grows, Bayesian optimization maintains near-optimal performance with logarithmically growing evaluation counts while grid search scales exponentially.

TABLE VIII
Search Strategy Comparison — Grid vs Random vs Bayesian

Job	Grid (20)	Random (10)	Bayesian (12)	Gap	Efficiency
Join	1.78s	1.81s	1.78s	0.0%	40% fewer evals*
Aggregation	8.09s	8.22s	8.09s	0.0%	40% fewer evals
Filter	6.09s	6.48s	6.09s	0.0%	40% fewer evals.
Window	8.47s	8.60s	8.47s	0.0%	40% fewer evals.

G. Temporal Stability Analysis

Table IX evaluates model consistency by splitting each job type's experiments into chronologically ordered early and late halves. Window and join workloads demonstrate strong temporal stability. The join model improves from R^2=0.655 to 0.944, suggesting that additional training data enhances prediction for complex multi-table operations.

TABLE IX
Temporal Stability Analysis — Early vs Late Experiment Halves

Job Type	Early R^2	Late R^2	Delta
Window	0.982	0.921	0.061
Join	0.655	0.944	0.289
Aggregation	0.844	0.121	0.723
Filter	0.040	0.196	0.156

H. Business Impact Analysis

Table X presents business impact projections using real AWS EMR pricing across three deployment scenarios, based on the observed average runtime reduction of 11.4%.

TABLE X
Business Impact Analysis — Real AWS EMR Pricing (2025)

Scenario	Monthly Savings	Annual Savings	
Small Analytics Team	₹41,000	₹5.0 Lakhs	5-
Mid-Size Bank	₹10.1 Lakhs	₹1.2 Crores	20
Enterprise Bank	₹2.04 Crores	₹24.4 Crores	10

Based on 11.4% average runtime reduction, $0.10–$2.00/hour/node AWS EMR pricing, ₹83.5/USD exchange rate.

VII. DISCUSSION

A. Why ML Outperforms AQE on Join and Window Workloads

AQE optimizes within a fixed initial shuffle partition count, dynamically coalescing partitions post-shuffle based on actual data statistics. For join-heavy banking analytics, the default 200 partitions creates excessive task scheduling overhead before AQE can intervene. The ML optimizer recommends shuffle=16 pre-execution, eliminating this overhead entirely. For window functions, optimal parallelism is determined by account cardinality — a relationship captured in the estimated_partitions and join_intensity features but inaccessible to AQE's runtime-only view.

B. The Banking-Specific fraud_complexity Feature

The fraud_complexity engineered feature (= has_fraud × (1 + num_joins)) addresses a banking-specific challenge: fraud queries scan skewed data distributions where 0.17% of rows carry the fraud flag. This creates uneven partition load not captured by standard size-based features. SHAP analysis confirms fraud_complexity contributes meaningfully across all job types, validating domain-specific feature engineering over generic Spark profiling approaches.

C. Limitations and Honest Assessment

Four limitations warrant acknowledgment. First, experiments are conducted in a single-node local[*] mode environment. The three optimized parameters are directly applicable to distributed deployments where their impact is amplified, but multi-node cluster validation represents an important future direction. Second, filter and aggregation workloads show prediction instability in late temporal periods ($R^2 < 0.20$), attributable to data-dependent execution paths where fraud predicate selectivity

varies across partitions. Third, the training dataset of 384 samples reflects controlled experimental conditions; production deployment benefits from continuous learning on real job histories. Fourth, at 25M rows the single-node memory constraint limits join optimization opportunities.

VIII. CONCLUSION

This paper presents SparkOptimizer-ML, an XGBoost-based system for pre-execution Spark configuration optimization on banking transaction workloads. The system achieves CV R^2=0.891 (±0.039) with log-transformed target encoding, improves 17 of 20 multi-scale evaluation scenarios, outperforms Apache Spark AQE by 41.9% on average across compute-intensive workloads, and delivers a 77.4% runtime reduction on window function queries. The Bayesian optimization component achieves 0% optimality gap with 40% fewer evaluations, and SHAP analysis provides workload-specific interpretability unavailable from black-box approaches. At enterprise scale, the system projects annual savings of ₹24 crores on 100-node AWS EMR deployments.

The central finding is that pre-execution ML-based parameter selection is complementary to, and in several workload categories superior to, Spark's built-in runtime adaptation. For the join and window workloads that dominate banking analytics pipelines, predicting optimal parameters before execution outperforms AQE's reactive optimization by substantial margins.

Future work will address current limitations: (1) validation on multi-node GCP Dataproc and AWS EMR clusters; (2) expansion to ten or more configuration parameters using Bayesian search to maintain evaluation efficiency; (3) online learning for continuous adaptation to workload evolution; (4) SHAP-guided per-workload feature selection to improve filter and aggregation prediction stability; and (5) integration with Spark's logical plan parser to enable automatic feature extraction from PySpark code without SQL string input.

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